

Stability and Performance Analysis for Non-Model-Based Friction Estimators

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Abstract

Many motion control systems use friction cancellation to improve precision. In non-model-based methods, well known rigid body dynamics and motion measurements are used to extract the unknown friction force acting on the system, i.e. friction is estimated without posing a structured nonlinear friction model. While the validity of non-model-based techniques for friction cancellation has been previously demonstrated experimentally, the stability of these methods remains to be shown. In this paper, the stability and experimental performance of three non non-model-based friction estimators are considered. These include a Kalman filter based approach, a predictive filter based approach and a local function estimation approach. Eigen analysis is used to determine the stability margins for the first two approaches, and Lyapunov analysis is used for the third. The analyses show that all three friction compensators are stable for a broad range of operating conditions and that experimental observations match theoretical predictions of stability margins. Experimental results show that the Kalman filter performs best overall, while the difficulty in tuning constrains local function estimation. The predictive filter friction estimate has undesirable high frequency content which when filtered leads to improved performance.

1 Introduction

Friction severely limits motion control efficiency in many mechanical systems. To reduce the need for high gain control laws, many systems are designed to estimate friction online in real time. The estimate is then added to the control input to 'cancel' friction directly. Such friction estimators are usually model-based, i.e., phenomenological or empirical modeling is used to characterize friction and online observers adaptively estimate the parameters of such models. As the friction process is more accurately modeled,

the cancellation is more successful and motion control efficiency is improved. [1] provides a comprehensive comparative study of some early model-based adaptive friction compensators and [2, 3] provide more recent examples of model-based friction compensation. In addition to model-based estimation, there are non-model-based approaches to estimating friction. Non-model-based friction estimation refers to estimating friction without attempting to model the complicated physical phenomena that lead to friction. Instead, well known system dynamics and motion measurements are used to back calculate friction. Non-model-based friction observers were considered in [4], which also compared model-based and non-model-based estimators in sinusoidal position tracking experiments. The results of this study indicate that non-model-based estimators are a viable alternative to model-based ones and in fact have superior performance for five out of six cases tested [4].

While the stability of model-based friction observers has been previously discussed by their respective proponents, a rigorous stability analysis of non-model-based friction observers has not been considered and is the primary objective of this paper. This paper considers the stability of three non-model-based friction estimators: a Kalman Filter (KF) based approach, a predictive filter (PF) based approach and local function estimation (LFE) based approach. In addition to theoretical stability analysis, this paper verifies the stability claims experimentally and provides performance comparisons for the three non-model-based observers in sinusoidal position tracking experiments. Like the stability analysis of model-based observers, examples of which are given in [2] and [3], stability of non-model-based observers is considered for a DC motor driven inertia, in which nonlinear friction behavior arises due to rolling and/or sliding contacts.

The Kalman filter appends the friction torque and its time derivatives as state elements using a random walk model and estimates friction torque along with the

other state elements. This state extended Kalman filter approach is summarized in [5]. The predictive filter [6] treats friction torque as the model error, which is estimated in real time. In local function estimation [7], an unknown nonlinear external input (which in this case is the friction torque) is estimated whenever a pre-specified position error tolerance is exceeded. Thus friction is not estimated at all times, but only locally.

Performance of the Kalman filter and local function estimation approaches was considered in [4]. Performance of the predictive filtering approach was evaluated for friction estimation in [8]. This paper is organized as follows. Section 2 briefly describes the experimental apparatus used. Section 3 presents the theoretical stability analysis for each of the non-model-based estimators, and sections 4 and 5 describe the experimental procedure and results.

2 Experimental facility and model

An apparatus was designed to test different friction cancellation schemes. It consists of a DC motor that drives a heavy disc (Figure 1). There are three bearings in the transmission path, which introduce nonlinear friction behavior. An optional cylindrical rider in contact with the disk and under static normal load provides an additional source of friction. The motivation behind using additional friction sources is to elicit a wide variety of friction behaviors.

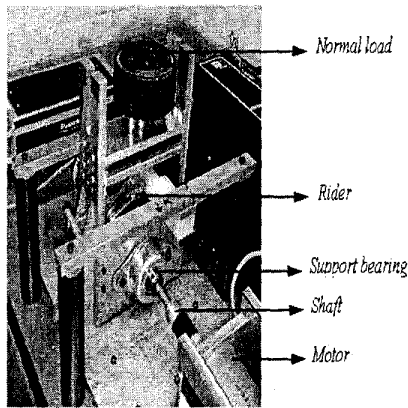


Figure 1: Experimental Apparatus

The apparatus is modeled as a second order system driven by a known input voltage V_{in} and influenced by an unknown external friction torque T_f

$$\begin{bmatrix} \ddot{\theta}(t) \\ \dot{\theta}(t) \end{bmatrix} = \begin{bmatrix} -B/J & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}(t) \\ \theta(t) \end{bmatrix} + \begin{bmatrix} K/J \\ 0 \end{bmatrix} V_{in} - \begin{bmatrix} 1/J \\ 0 \end{bmatrix} T_f(t) \quad (1)$$

where $\dot{\theta}$ and θ are the shaft angular velocity and position, J is the moment of inertia of the disk and motor, B is the viscous damping in the bearings and K is a motor constant. The motor is driven by a linear amplifier in torque control mode. An optical encoder with a resolution of 6.34×10^{-4} radians is used to provide accurate position measurements.

Position tracking experiments are carried out in which a well tuned PID compensator controls the position of the disc to follow a sinusoidal trajectory (20 degrees peak to peak at 0.1Hz, 0.3Hz and 1.0Hz). The bandwidth of the PID compensated system is 10 Hz, which is the largest bandwidth achievable without compromising stability. Although the PID compensated system achieves tracking errors of the order of 10^{-4} radians in friction free simulations, in practice the tracking errors are much higher due to friction. This is especially true when the line contact is exercised with a 5 lb normal load as shown in Figure 1. The net control input to the motor therefore consists of the input from the PID controller plus the estimated friction. A feed-forward acceleration term is also added.

Figure 2 shows a block diagram of the friction compensator. One of the goals of this research is to estimate the gain margin (M) of the friction canceled system. As shown in Figure 2, the friction estimate is multiplied by M before being used for cancellation. $M = 0$ is a baseline PID compensated system with no friction cancellation while $M = 1$ indicates 'exact' friction cancellation. It is exact in the sense that if the friction torque estimate ($\hat{T}_f$) is exactly equal to the true friction torque (T_f) then $\hat{T}_f - T_f = 0$. If the system is stable for $M > 1$ it means that there is a positive gain margin and some gain uncertainty can be tolerated in the friction observer.

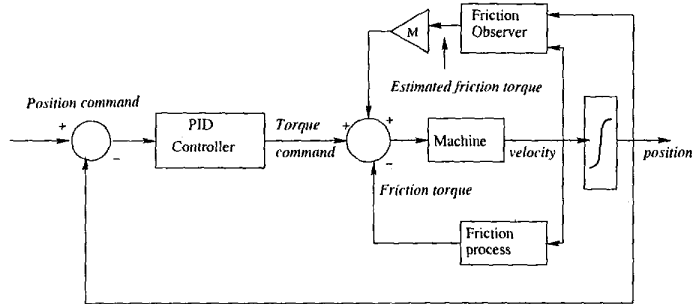


Figure 2: Block diagram of friction compensated system

Numerical values used in (1) for the apparatus are $J = 0.0056 \text{ Kg-m}^2$, $B = 0.0067 \text{ kg-m}^2/\text{s}$ and $K = 54.04 \text{ Nm/V}$. Additional details regarding the apparatus are provided in [4].

3 Stability analysis of non-model-based compensators

3.1 Kalman filter based friction estimation

In this method, friction torque is appended to the system state using a random walk model and estimated along with the other state elements using the standard state extended Kalman observer. The first order random walk model ($\dot{T}_f = 0$) treats the friction torque as a random constant within each sampling instant. The second order random walk model, which is assumed in the following analysis treats friction torque as a linear function of time. Note that the term *random walk model* is not to be confused with the structured *friction models* used in model-based-observers. Used here, it is simply a state augmentation technique. The entries in the process noise covariance matrix (Q) corresponding to the appended friction state and its derivative become the tuning parameters.

The state and measurement equations of the Kalman filter are,

$$x_k = \Phi x_{k-1} + \Gamma u_k + \Lambda w_k \quad (2)$$

$$z_k = H x_k + n_k \quad (3)$$

where Φ, Γ, Λ are the state transition, control effect and process noise input matrices. For the second order random walk model, the state is $[\dot{\theta}_k, \theta_k, T_{fk}, \dot{T}_{fk}]^T$. Q is the covariance matrix of the process noise, w_k . n_k is the measurement noise and u_k is the net input to the system. H is the output matrix.

The Kalman filter based state estimate is given by,

$$\hat{x}_k = \Phi \hat{x}_{k-1} + \Gamma u_k + K_k \{z_k - H \Phi \hat{x}_{k-1} - H \Gamma u_k\} \quad (4)$$

where K_k is a vector of Kalman gains. The stability of the Kalman filter as an *open loop* state estimator is well established and the derivation can be found in many standard texts such as [5]. We are however concerned with the *closed loop* friction compensated system. For this system, the state vector consists of the shaft velocity and position, the friction torque estimate and its derivative. We therefore represent the input as,

$$u_k = u_{PIDk} + u_{FFk} + M \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \hat{x}_{k-1} \quad (5)$$

where u_{PIDk} is the control input from the PID controller, u_{FFk} is the feed-forward acceleration term and M is the friction feedback amplification factor, which is equivalent to the gain margin of the closed loop system. One of the aims of this analysis is to find the range of M for which the closed loop system is stable. The matrix $\begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}$ is so chosen for friction cancel-

lation using the estimate $\hat{T}_f$. Using (5) in (4) we have,

$$\begin{aligned} \hat{x}_k = & \{\Phi + \Gamma M \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} - K_k H \Phi \\ & - K_k H \Gamma M \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}\} \hat{x}_{k-1} \\ & + \{\Gamma - K_k H \Gamma\} (u_{PIDk} + u_{FFk}) + K_k z_k \end{aligned} \quad (6)$$

which can be re-written as,

$$\begin{aligned} \hat{x}_k = & \Phi' \hat{x}_{k-1} + \Gamma' (u_{PIDk} + u_{FFk}) + K_k z_k \\ \Phi' = & \Phi + \Gamma M \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} - K_k H \Phi - K_k H \Gamma M \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} \\ \Gamma' = & \Gamma - K_k H \Gamma \end{aligned} \quad (7)$$

Since u_{PIDk} and u_{FFk} are external inputs and z_k is the measurement, instability results when the eigenvalues of Φ' fall outside the unit circle. This allows us to determine the range of M for which the system is stable. Although the Kalman gain matrix is indexed with k , in practice it converges very quickly and remains constant, providing stability analysis at steady state.

Using the numerical values from section 2, it is found that the experimental system is stable for $0 \leq M \leq 1.4$. This analysis predicts that the stability properties of the Kalman filter based friction compensator are independent of the operating frequency and loading conditions. Similar results are obtained when this analysis was repeated for a third order random walk model.

3.2 Predictive filter based friction estimation

The predictive filter estimates the *model error* in real time. When applied to friction cancellation, the friction torque is considered as the model error and estimated online. The estimator is given by,

$$\hat{x}_{k+1} = \Phi \hat{x}_k + G d_k + \Gamma u_k \quad (8)$$

where the state vector is $x_k = [x_{1k}, x_{2k}]^T = [\dot{\theta}_k, \theta_k]^T$. G is the model error distribution matrix and u_k is again the total input voltage to the system. The analysis outlined in [6] yields the following expression for the model error d_k . Details of this derivation are found in [4].

$$d_k = Q (2x_{1k} \Delta + a \Delta^2 x_{1k} - 2y_{k+1} + 2x_{2k}) \quad (9)$$

where $Q = \frac{-\Delta^2 c}{4\Delta^4 c^2 + 4\omega r}$ and $\Delta = 0.01$ is the sample time and the constants are $a = B/J$ and $c = 1/J$. ω is a tuning parameter which needs to be tuned for optimal performance. y_{k+1} is considered the current position measurement.

The stability of the predictive filter as an open loop estimator is well established and the proof is found in [9]. Considered here is the *closed loop* stability of the predictive filter based friction compensator. In the closed loop system, the estimated friction torque, d_k forms part of the input:

$$u_k = u_{PIDk} + u_{FFk} + M d_k \quad (10)$$

where as before, u_{PIDk} is the control input from the PID controller, u_{FF} is the feedforward acceleration term and M is the friction feedback amplification factor. As before, one of the aims of stability analysis is to find a range of M for which the closed loop system is stable. Using (10) in (8) yields,

$$\hat{x}_{k+1} = \Phi \hat{x}_k + Gd_k + \Gamma(u_{PIDk} + u_{FFk}) + \Gamma M d_k \quad (11)$$

using (9), we can write d_k as a function of the state and the measurement:

$$\begin{aligned} Gd_k &= \begin{bmatrix} g_{11} \\ g_{21} \end{bmatrix} d_k \\ &= \begin{bmatrix} Qg_{11}(2\Delta + a\Delta^2) & 2Qg_{11} \\ Qg_{21}(2\Delta + a\Delta^2) & 2Qg_{21} \end{bmatrix} \begin{bmatrix} x_{1k} \\ x_{2k} \end{bmatrix} \\ &\quad + \begin{bmatrix} -2Qg_{11} \\ -2Qg_{21} \end{bmatrix} y_{k+1} \\ &= Rx_k + Ty_{k+1} \end{aligned} \quad (12)$$

with

$$\begin{aligned} R &= \begin{bmatrix} Qg_{11}(2\Delta + a\Delta^2) & 2Qg_{11} \\ Qg_{21}(2\Delta + a\Delta^2) & 2Qg_{21} \end{bmatrix} \\ T &= \begin{bmatrix} -2Qg_{11} \\ -2Qg_{21} \end{bmatrix} \end{aligned} \quad (13)$$

Similarly, $\Gamma M d_k$ may be written as

$$\Gamma M d_k = M \begin{bmatrix} \Gamma_{11} \\ \Gamma_{21} \end{bmatrix} d_k = Dx_k + Ey_{k+1} \quad (14)$$

with

$$\begin{aligned} D &= \begin{bmatrix} MQ\Gamma_{11}(2\Delta + a\Delta^2) & 2MQ\Gamma_{11} \\ MQ\Gamma_{21}(2\Delta + a\Delta^2) & 2MQ\Gamma_{21} \end{bmatrix} \\ E &= \begin{bmatrix} -2MQ\Gamma_{11} \\ -2MQ\Gamma_{21} \end{bmatrix} \end{aligned} \quad (15)$$

Substituting (12) and (14) in (11) we have,

$$\begin{aligned} \hat{x}_{k+1} &= (\Phi + R + D) \hat{x}_k + (T + E) y_{k+1} \\ &\quad + \Gamma(u_{PIDk} + u_{FFk}) \end{aligned} \quad (16)$$

Since u_{PID} and u_{FF} are inputs external to the predictive filter and y_{k+1} is a measurement, we can analyze stability by looking at the eigenvalues of $\Phi + R + D$ as functions of M . Using numerical values in (16), it was found that the system is stable for $0 \leq M \leq 1.5$. As in the Kalman Filter, it is observed that the stability characteristics are independent of driving frequency and loading condition.

3.3 Friction Estimation Using Local Function Estimation

Local function estimation [7] is based on Model Referenced Adaptive Control (MRAC). The plant and the reference model are given by,

$$\dot{x} = Ax + Bu + f(x) \quad (17)$$

$$\dot{x}_m = A_m x_m + B_m r \quad (18)$$

where $[A, B]$ and $[A_m, B_m]$ are the plant and the reference model in state-space form. r is a reference input and $f(x) = [f_1(x), f_2(x), \dots, f_n(x)]^T$ is an unknown possibly nonlinear vector function of the system state. In this paper, it is the friction torque. The input u for the MRAC controller is given by,

$$u = -Kx + \Delta r + u_{ad} \quad (19)$$

where $A - BK = AM$ and $B\Delta = BM$. u_{ad} is an adaptive term that cancels $f_1(x, x_i)$, the first order Taylor series approximation of $f(x)$ at x_i . The adaptation law works as follows. An n -dimensional tolerance sphere is defined around the reference trajectory:

$$\Phi(t, \rho) = \{x \in R^n \mid \|x(t) - x_m(t)\| \leq \rho\} \quad (20)$$

The first time the state leaves the tolerance sphere is sampled as x_1 . The second time it leaves the sphere it is sampled as x_2 and so on. At each x_i we estimate and cancel the first order Taylor series expansion of $f(x)$ at x_i . In essence ρ represents tolerable error, which is reduced by friction cancellation. The adaptation law is therefore given by,

$$\begin{aligned} u_{ad} &= -\hat{f}_1(x, x_i) = \hat{C}_0(x_i, t) + \hat{C}_1(x_i, t)(x - x_i) \\ \|x - x_m\| &> \rho \\ u_{ad} &= 0, \quad \|x - x_m\| \leq \rho \end{aligned} \quad (21)$$

where the C_k matrices are given by,

$$\dot{\hat{C}}_k(x_i, t) = -\gamma B^T P e w_k(x, x_i) \quad (22)$$

$w_0(x, x_i) = [1, \dots, 1]^T$ and $w_1(x, x_i) = (x - x_i)$ and $e = x - x_m$ is the state error. γ is a tuning parameter. P is the solution to the matrix Lyapunov equation, $A_m^T P + P A_m = -I$.

The Kalman filter and the predictive filter were originally presented as open loop state estimators. The open loop stability of these estimators is well established. This research makes the additional step of using that estimate to cancel friction, hence closing the loop. Sections 3.1 and 3.2 provide the closed loop stability when the estimated friction is used to provide friction cancellation. Local function estimation however is presented as a closed loop system. It is based on estimating and canceling an unknown nonlinear function (i.e. friction) locally and is inherently closed loop. For this reason, the stability analysis discussed in the original paper is valid in our context and is outlined here. The reader is referred to [7] for details.

To prove stability, $f_i(x, x_i)$ is expressed as

$$\begin{aligned} f_i(x, x_i) &= \sum_{k=0}^{k=1} G_k(x_i) w_k(x, x_i) \\ &= -B \sum_{k=0}^{k=1} C_k(x_i) w_k(x, x_i) \end{aligned} \quad (23)$$

where $G_0(x) = \text{diag}[f_1(x), \dots, f_n(x)]$ and $[G_1(x)]_{ij} = [\partial f_i(x)/\partial x_j]$ are the terms in the Taylor series expansion. The following Lyapunov function is used to prove stability:

$$V(x_i, t) = \frac{1}{2}e^T P e + \frac{1}{2\gamma} \sum_{k=0}^{k=1} \text{Tr} [\tilde{C}_k^T \tilde{C}_k] \quad (24)$$

with $\tilde{C}_k(x_i) = \hat{C}_k(x_i) - C_k(x_i)$ for $k = 0, 1$ and $\text{Tr}[\cdot]$ is matrix trace. $\dot{V}$, the time derivative of the Lyapunov function along the system trajectory is calculated as,

$$\begin{aligned} \dot{V} = & -\frac{1}{2}e^T P e + e^T P f_e(x, x_i) \\ & + e^T P B \sum_{k=0}^{k=1} \tilde{C}_k w_k(x, x_i) + \frac{1}{\gamma} \sum_{k=0}^{k=1} \text{Tr} [\dot{\tilde{C}}_k^T \tilde{C}_k] \end{aligned} \quad (25)$$

where $f_e(x, x_i) = f(x) - f_1(x, x_i)$ is the error incurred by using the Taylor expansion. Using (22) in (25) yields,

$$\dot{V} = -\frac{1}{2}e^T P e + e^T P f_e(x, x_i) \quad (26)$$

Now for whatever $\rho > 0$ we choose, $\exists \epsilon$ such that $\|f_e(x, x_i)\| \leq \epsilon = \frac{\rho}{4\|P\|}$ for $\|e\| \geq \rho$ Using these in (26) we get,

$$\begin{aligned} \dot{V} & \leq -\frac{1}{2}\|e\|^2 + \|P\| \cdot \|e\| \cdot \|f_e(x, x_i)\| \\ & \leq -\frac{1}{2}\|e\|^2 + \frac{\rho}{4}\|e\| \\ & \leq -\frac{\rho^2}{4} \end{aligned} \quad (27)$$

for $\|e\| \geq \rho$, thus proving stability. Using (27) it can be shown that when the error tolerance is exceeded, the state must return to within tolerance in finite time. Proof of this theorem and more details are found in [7].

4 Experimental Procedure

Performance of each friction compensator is examined using the experimental apparatus of Figure 1, in which the rotating disk must track a sinusoidal reference input, $\theta_d(t) = A \sin(2\pi f t)$ for three different frequencies, $f = 0.1\text{Hz}$, 0.3Hz and 1.0Hz . $A = 0.1745$, which denotes 20 degrees peak to peak angular displacement of the disk. At 0.1Hz , motion is stick slip dominated and static friction and Stribeck effect are pronounced. Coulomb friction dominates at higher frequencies. All the results in this paper were obtained with the 5 lb normal load exercising the line contact as shown in Figure 1. As mentioned in section 2, each friction compensator is used in combination with a well tuned PID controller to reduce the root mean squared position tracking error (RMSE), which is used as a performance metric. Two sets of data were taken for each compensator.

Four complete cycles of data were taken for the 0.1Hz input (40 seconds of data), 6.6 cycles for the 0.3Hz input (20 seconds of data) and 10 cycles of data for the 1.0Hz input (10 seconds of data). All experiments were carried out in real time using Simulink/RTW in a dSPACE environment with a sample time of 0.01s .

Each compensator has one or more tuning parameters. All tuning was done for the 0.3Hz input and fixed for the other two inputs. Tuning parameters were not altered for the other frequencies even though such alteration can lead to better performance. The tuned parameters are summarized in Table 1. All three compensators were tuned through trial and error with the goal of minimizing the RMSE for the 0.3Hz input. In a comparative study of model-based and non-model-based friction compensators [4], which used the same equipment to perform similar experiments, accurate friction measurements were available for comparison with the estimated friction torque. Thus parameters were also tuned to make the estimated friction match the expected friction torque as much as possible.

Table 1: Experimentally determined parameters for each adaptive friction compensation algorithm

Method	Parameters
KF	$Q_{\dot{\theta}} = Q_{\theta} = 0; Q_{Tf} = Q_{Tf} = 2e - 4$
PF	$w = 2.5$
LFE	$\gamma = 15.0$

Of all the friction compensators considered in this paper and in [4], the Kalman filter was the easiest to tune. For the predictive filter, optimal ω can be anywhere in $(0, \infty)$. However tuning is relatively simple once the general range is established. Local function estimation is the hardest to tune and friction estimate magnitude is very sensitive to γ .

The analysis presented in section 2 neglects any higher-order dynamics of the experimental system. In practice, all three non-model-based observers produce friction torque estimates with substantial power at high frequencies. Of the three, the predictive filter was most susceptible to high frequency content in the friction torque estimate. It was found that due to this, the predictive filter friction estimate was unsuited to provide friction cancellation. The high frequency content quickly introduced instability. An 8th order IIR low pass filter was used to attenuate the higher frequencies in order to stabilize the friction cancellation path in the face of unmodeled system dynamics.

5 Experimental Results

Table 2 summarizes mean RMS tracking errors for each friction compensation algorithm and driving frequency based on two trials. It also includes the RMS errors obtained with the baseline PID compensator without any friction cancellation.

Table 2: Root mean squared errors (in radians) from position tracking experiments

Method	$f = 0.1\text{Hz}$	$f = 0.3\text{Hz}$	$f = 1.0\text{Hz}$
PID	0.0122	0.0270	0.0471
KF	0.0041	0.0101	0.0273
PF	0.0073	0.0185	0.0444
LFE	0.0117	0.0243	0.0690

It is readily observed from Table 2 that each friction compensator provides some reduction in the RMS tracking error over the baseline PID case. The Kalman filter provides the best friction cancellation overall and it reduces the RMS error by a factor of two to three. For the 0.1Hz case, the RMS error for the KF observer is less than a factor of 10 greater than the encoder resolution, indicating the nearness to the best possible friction cancellation results obtainable with the present apparatus. The predictive filter has an acceptable performance overall and provides moderate reduction in RMS tracking errors for all three frequencies. Local function estimation overcompensates for friction for the 1.0Hz and only marginally improves the performance for the other two cases.

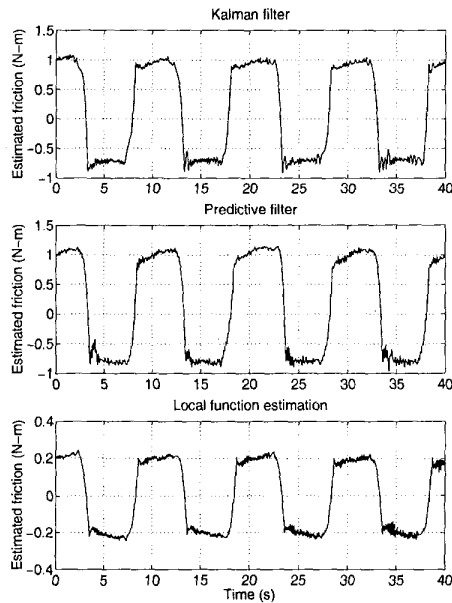


Figure 3: Friction estimate obtained from each of the methods for the 0.1Hz input

Figure 3 shows the friction estimates obtained using the three methods for the 0.1Hz input. First it is noted that all three friction estimates match in shape. The friction estimate magnitudes for the Kalman filter and the predictive filter agree with the friction measurements reported in [4], which used the same equipment. While the Kalman filter and the predictive filter agree in magnitude, the local function estimation is off by a factor of four. This is attributed to the difficulty in tuning the LFE. For the value of γ reported in Table 1, the friction estimate *shape* obtained with LFE agrees with the other two methods. The shape also agrees with the friction measurements reported in [4]. Increasing γ results in a corresponding increase in the friction magnitude, but the shape of the estimate is no longer accurate and this inaccurate estimate actually increases the tracking error instead of reducing it. This is shown in Figure 4 for the 0.1Hz input. When γ is increased to fix the magnitude, the tracking RMS error (0.0175 radians) is higher than that obtained with no friction cancellation (0.0122 radians).

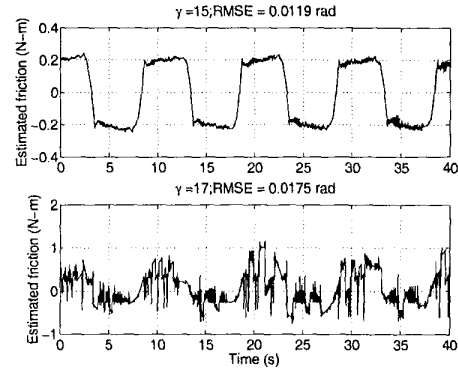


Figure 4: Friction estimates obtained using local function estimation for two different settings of γ for the 0.1Hz input

Figure 5 shows the effect of increasing M , the friction feedback amplification factor or gain margin of the system. When the system is stable for $M > 1$ it means that it has a positive gain margin and can tolerate gain uncertainty. The Kalman filter behaves as predicted by theory and becomes unstable for $M > 1.4$. The RMS tracking error increases dramatically at $M = 1.3$ indicating the onset of instability. Theoretical calculations indicate that the predictive filter should become unstable for $M > 1.5$. Experimentally it was observed that the system became unstable for $M > 1.7$. This discrepancy is attributed to the low pass filter that was applied to the predictive filter friction estimate prior to cancellation. The filter was not used in the theoretical calculations.

Though the PF based friction cancellation scheme has a higher gain margin than the KF based scheme, the RMS error for the PF is still higher than KF except

when the KF is close to instability. Although Figure 5 shows the results for a specific driving frequency of 0.3Hz, the stability margins did not change for the other two driving frequencies. This matches the theoretical predictions.

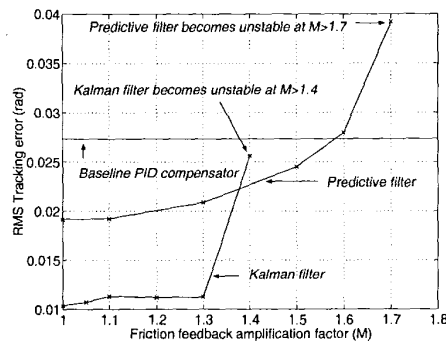


Figure 5: RMS tracking errors as functions of M for the 0.3Hz input

6 Conclusions

Two conclusions are drawn from this study. First non-model-based estimation is a stable and reliable alternative to model-based estimation of friction torque. The three non-model-based observers considered in this study are found to be stable for a wide range of friction behaviors and driving frequencies for a rigid motor driven inertia apparatus. The Kalman filter has the best performance over all and has a gain margin of 1.4.

Finally it is noted that tuning is a very important factor in selecting and designing adaptive control laws. The serious difficulties encountered in tuning the local function estimation resulted in a significant reduction in the performance of this elegant adaptive friction cancellation method. On the other hand, it was found that the Kalman filter was very easy to tune and performed well even in the face of modeling uncertainty. This makes the Kalman filter approach to friction cancellation a logical choice in practical applications.

Acknowledgment

This work was supported by the National Science Foundation under Contract No. CMS9721468. Their support is gratefully acknowledged.

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